Eric Wu

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Education

University of Waterloo **Sep. 2025 – Apr. 2030 (Expected)**
B.ASc in Mechatronics Engineering *Waterloo, ON*

Experience

Undergraduate Researcher **Sep. 2026 – Present**
University of Waterloo – SmartTatami *Waterloo, ON*

- Developing a scalable **Velostat pressure-sensing array** to measure contact location and relative force during judo throws, complementing multi-camera pose estimation.
- Designing a **custom PCB** in **Altium**, integrating an MCU, multiplexers, ADC, and signal-conditioning circuitry for pressure data acquisition.
- Designing **Pico firmware** to scan the array and **PLC ladder logic** to detect occupied zones from readings received over Modbus RTU.

Electromechanical Engineering Intern **May 2026 – Aug. 2026**
Spyder Controls *Lacombe, AB*

- Owned the end-to-end development of PCB **test fixtures**, from **AutoCAD** design and **GD&T** to fabrication, validation, and production deployment.
- Designed and deployed a **spring-loaded** PCB locating fixture for automatic board alignment, increasing testing throughput by **50%** to 400+ boards/day.
- Evaluated **electrical components** using **oscilloscopes, multimeters, and power supplies**, analysing test results for accuracy and thermal performance to guide component selection.
- Designed, fabricated and installed **copper shielding** in high-density PCB test fixtures, analysing test results to reduce cross-talk and false failures by **90%**.
- Root-caused electrical faults and restored **5 test fixtures** using **CAN bus**, electrical measurements, and wiring repairs.

Humanoid Robotics Engineer **Jan. 2026 – Present**
WATonomous *Waterloo, ON*

- Designed and prototyped **bipedal humanoid leg linkages** in **SolidWorks and Onshape**, reducing part count by **66%**.
- Implemented a **ROS 2 costmap** using simulated LiDAR data, tuning map resolution and obstacle inflation for collision-free navigation.
- Built a custom **A* pathfinding algorithm** to generate routes through the occupancy grid, then containerized the navigation stack in **Docker** and visualized paths in **Foxglove**.

Robotics Team Lead **Sep. 2022 – May 2025**
Churchill Robotics - VEX Team 3388C *Calgary, AB*

- Led design of 12+ competition robots, modelling drivetrains, scoring, and climbing mechanisms in **Fusion 360**.
- Created full robot CAD assemblies and custom laser-cut plates and brackets in **SolidWorks** and **Fusion 360**, iterating designs for fit, weight, and ease of fabrication.
- Achieved 1st and 3rd place finishes at the 2024 and 2025 Alberta Provincial Championships, with a top-20 national finish and top 2% overall among 20,000+ teams.

Projects

Omni-Assist | *Self-Driving Chair* **Apr. 2026**
1st Place in Hardware & Best Use of Viam, YHack @ Yale University

- Designed a low-cost electromechanical drivetrain that converts any four-legged chair into a self-navigating wheelchair.
- Integrated four DC gear motors, motor drivers, battery power, and voltage regulation into the drivetrain, controlled through an **Arduino**.
- Modelled ratcheting leg clamps, motor mounts, spacers, and wheel hubs in **SolidWorks**, then 3D-printed the components within 24 hours.
- Implemented **LiDAR** occupancy-grid mapping, A* path planning, and PID motion control for autonomous navigation and obstacle avoidance in C++.

Skills

ME: SolidWorks (CSWA Certified), Fusion 360, AutoCAD, Onshape, CorelDRAW, GD&T, Lathe, Mill, 3D Printing
EE: Altium, ICT Testing, Prototyping & Debugging, Soldering & Harnessing, Sensors, Power Electronics, CANbus, Oscilloscopes
SWE/Robotics: C++, Python, ROS 2, Linux, Git, STM32, Arduino, I^2C, LiDAR, PLC Ladder Logic